

Chap. 4. The Continuous-time Fourier Transform

- Continuous-Time Aperiodic Signals
- Continuous-Time Fourier Transform (CTFT)
- Convolution Property
- Multiplication Property

- Continuous-Time Fourier Series (CTFS)
- Discrete-Time Fourier Series (DTFS)
- Discrete-Time Fourier Transform (DTFT)
- Discrete Fourier Transform (DFT)
- Fast Fourier Transform (FFT)

§4.1 Representation of Aperiodic Signals:

The Continuous Time Fourier Transform (CTFT)

the period of the periodic signal : finite T

the period of the aperiodic signal : infinite T ($T \rightarrow \infty$)

(Example 3.5) a periodic square wave with period T

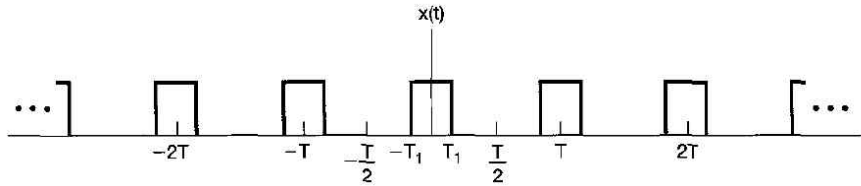


Figure 4.1 A continuous-time periodic square wave.

Fourier series coefficient a_k [Eq.(3.44)]

$$a_k = \frac{2 \sin(k \omega_0 T_1)}{k \omega_0 T} \quad (4.1)$$

$T = 4 T_1$ (with T_1 fixed)	(a)
$T = 8 T_1$ (with T_1 fixed)	(b)
$T = 16 T_1$ (with T_1 fixed)	(c)

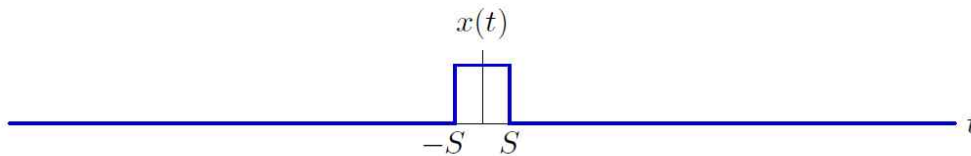
※ The values of the horizontal axis are ω , not k .

※ If $T \rightarrow \infty$, then CTFS $\rightarrow$ CTFT.

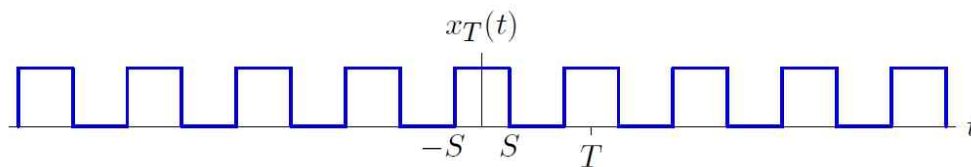
※ MIT OCW Lecture 16

An aperiodic signal can be thought of as periodic with infinite period.

Let $x(t)$ represent an aperiodic signal.

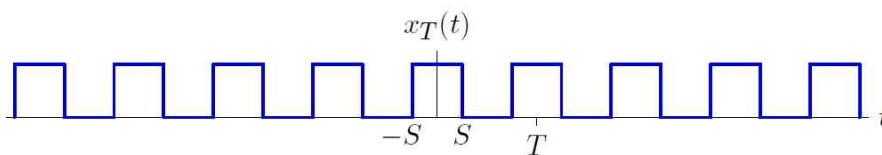


"Periodic extension":
$$x_T(t) = \sum_{k=-\infty}^{\infty} x(t + kT)$$

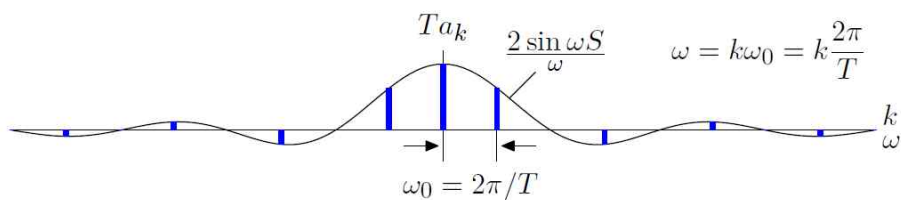


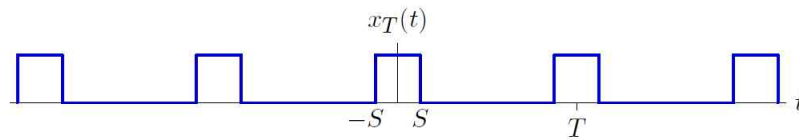
Then $x(t) = \lim_{T \rightarrow \infty} x_T(t)$.

Represent $x_T(t)$ by its Fourier series.

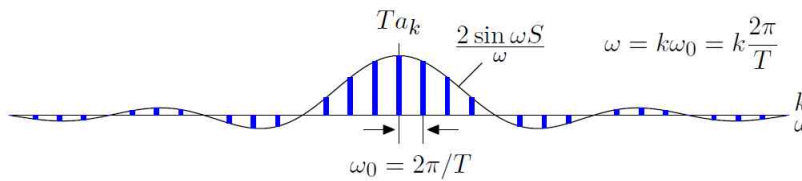


$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} x_T(t) e^{-j\frac{2\pi}{T}kt} dt = \frac{1}{T} \int_{-S}^S e^{-j\frac{2\pi}{T}kt} dt = \frac{\sin \frac{2\pi kS}{T}}{\pi k} = \frac{2}{T} \frac{\sin \omega S}{\omega}$$

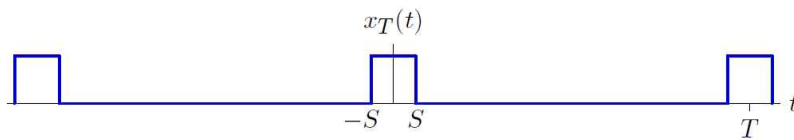


Period (T) doubles, pulse width (τ) fixed

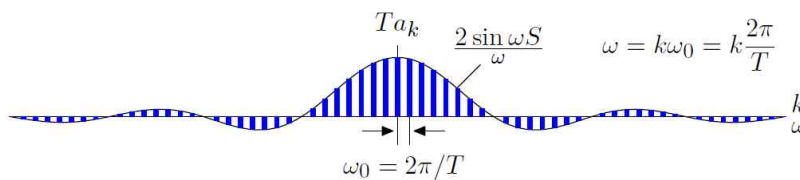
$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} x_T(t) e^{-j\frac{2\pi}{T}kt} dt = \frac{1}{T} \int_{-S}^S e^{-j\frac{2\pi}{T}kt} dt = \frac{\sin \frac{2\pi kS}{T}}{\pi k} = \frac{2}{T} \frac{\sin \omega S}{\omega}$$



T bigger and bigger, ω_0 (or $f_0 = \frac{1}{T}$) smaller and smaller)

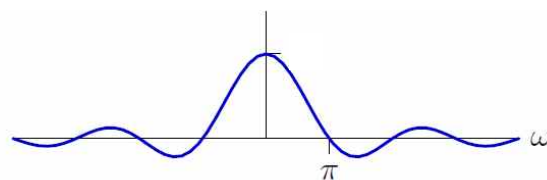
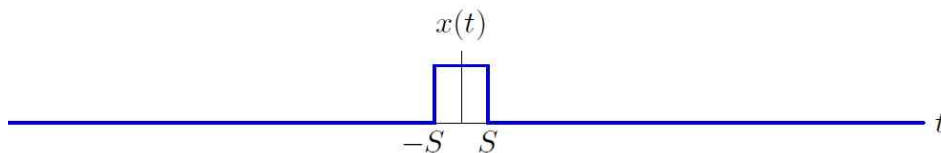


$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} x_T(t) e^{-j\frac{2\pi}{T}kt} dt = \frac{1}{T} \int_{-S}^S e^{-j\frac{2\pi}{T}kt} dt = \frac{\sin \frac{2\pi kS}{T}}{\pi k} = \frac{2}{T} \frac{\sin \omega S}{\omega}$$



$$\lim_{T \rightarrow \infty} T a_k = \lim_{T \rightarrow \infty} \int_{-T/2}^{T/2} x(t) e^{-j\omega t} dt = \frac{2}{\omega} \sin \omega S = E(\omega)$$

$T \rightarrow \infty$ (nonperiodic function), ω_0 (or $f_0 = \frac{1}{T}$) $\rightarrow 0$



$$X(j\omega) = \frac{2 \sin (\omega)}{\omega}$$

● CTFS $\Rightarrow$ CTFT

continuous-time periodic signal $\tilde{x}(t)$	$T \rightarrow \infty$ $\omega_0 \rightarrow 0$	continuous-time nonperiodic signal $x(t)$
Continuous-Time Fourier Series a_k (CTFS)		Continuous-Time Fourier Transform (CTFT)
$\tilde{x}(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$ $= \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t}$		$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$
$a_k = \frac{1}{T} \int_T \tilde{x}(t) e^{-jk\omega_0 t} dt$ $= \frac{1}{T} \int_T \tilde{x}(t) e^{-jk(2\pi/T)t} dt$		$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$

Since $\tilde{x}(t) = x(t)$ for $|t| < T/2$, then

$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_{-\infty}^{\infty} x(t) e^{-jk\omega_0 t} dt$$

Defining the envelope $X(j\omega)$ of Ta_k

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

- Relationship between a_k and $X(j\omega)$:

$$a_k = \frac{1}{T} X(jk\omega_0)$$

Aperiodic signal $\tilde{x}(t)$ can be expressed as $X(j\omega)$

$$\tilde{x}(t) = \sum_{k=-\infty}^{+\infty} \frac{1}{T} X(jk\omega_0) e^{jk\omega_0 t} = \frac{1}{2\pi} \sum_{k=-\infty}^{+\infty} X(jk\omega_0) e^{jk\omega_0 t} \omega_0$$

● Continuous-Time Fourier Transform (CTFT) Pair

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \quad (4.8)$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad (4.9)$$

※ $X(j\omega)$ is the spectrum of an aperiodic signal $x(t)$.

§4.1.2 Convergence of Fourier Transform

If $x(t)$ has finite energy, that is, $\int_{-\infty}^{\infty} |x(t)|^2 dt < \infty$,

then we are guaranteed that $X(j\omega)$ is finite
[i.e., Eq.(4.9) converges].

● Dirichlet conditions

1. $x(t)$ must be absolutely integrable, that is,

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty.$$

2. $x(t)$ has a finite number of maxima and minima within any finite interval.
3. $x(t)$ has a finite number of discontinuities within any finite interval. Furthermore, each of these discontinuities must be finite.

● Relation between Fourier Transform and Laplace Transform

If the Laplace transform of a signal exists and if the ROC includes the $j\omega$ axis, then the Fourier transform is equal to the Laplace transform evaluated on the $j\omega$ axis.

Laplace transform:

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st} dt$$

Fourier transform:

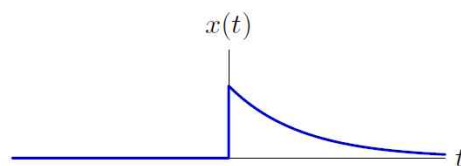
$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt = X(s)|_{s=j\omega}$$

Fourier transform “inherits” properties of Laplace transform.

Property	$x(t)$	$X(s)$	$X(j\omega)$
Linearity	$ax_1(t) + bx_2(t)$	$aX_1(s) + bX_2(s)$	$aX_1(j\omega) + bX_2(j\omega)$
Time shift	$x(t - t_0)$	$e^{-st_0}X(s)$	$e^{-j\omega t_0}X(j\omega)$
Time scale	$x(at)$	$\frac{1}{ a }X\left(\frac{s}{a}\right)$	$\frac{1}{ a }X\left(\frac{j\omega}{a}\right)$
Differentiation	$\frac{dx(t)}{dt}$	$sX(s)$	$j\omega X(j\omega)$
Multiply by t	$tx(t)$	$-\frac{d}{ds}X(s)$	$-\frac{1}{j}\frac{d}{d\omega}X(j\omega)$
Convolution	$x_1(t) * x_2(t)$	$X_1(s) \times X_2(s)$	$X_1(j\omega) \times X_2(j\omega)$

There are also important differences.

Compare Fourier and Laplace transforms of $x(t) = e^{-t}u(t)$.



Laplace transform

$$X(s) = \int_{-\infty}^{\infty} e^{-t}u(t)e^{-st} dt = \int_0^{\infty} e^{-(s+1)t} dt = \frac{1}{1+s} ; \text{Re}(s) > -1$$

a complex-valued function of **complex** domain.

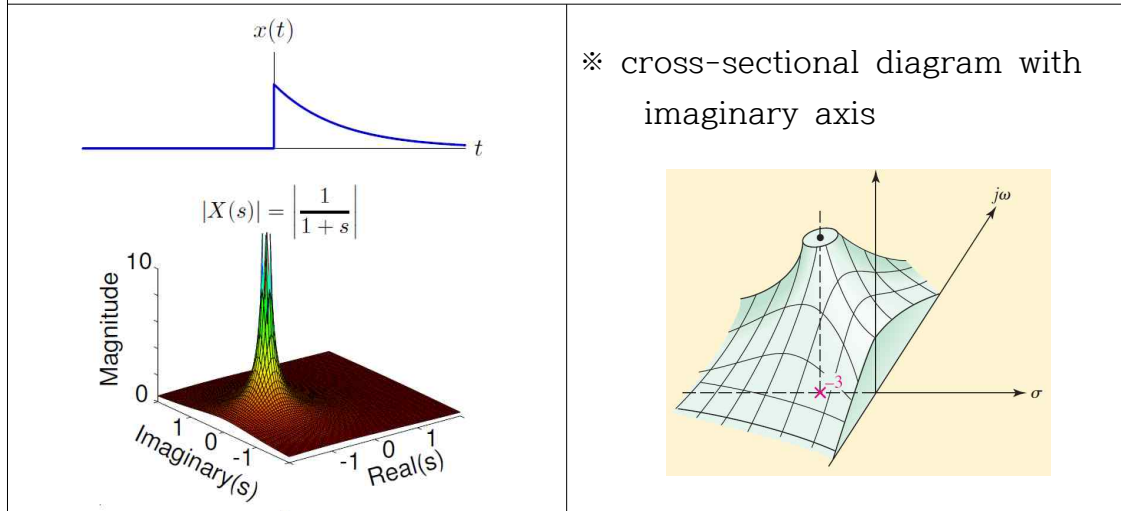
Fourier transform

$$X(j\omega) = \int_{-\infty}^{\infty} e^{-t}u(t)e^{-j\omega t} dt = \int_0^{\infty} e^{-(j\omega+1)t} dt = \frac{1}{1+j\omega}$$

a complex-valued function of **real** domain.

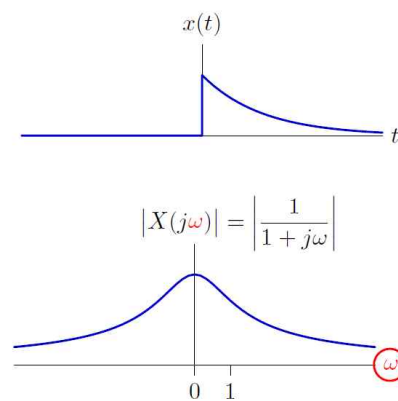
※ Laplace Transform

The Laplace transform maps a function of time t to a complex-valued function of complex-valued domain s .



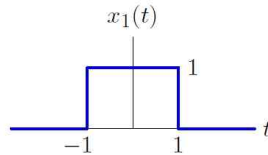
※ Fourier Transform

The Fourier transform maps a function of time t to a complex-valued function of real-valued domain ω .



Frequency plots provide intuition that is difficult to otherwise obtain.

※ LT and FT of a square pulse



Laplace transform:

$$X_1(s) = \int_{-1}^1 e^{-st} dt = \left. \frac{e^{-st}}{-s} \right|_{-1}^1 = \frac{1}{s} e^s - e^{-s} \quad [\text{function of } s = \sigma + j\omega]$$

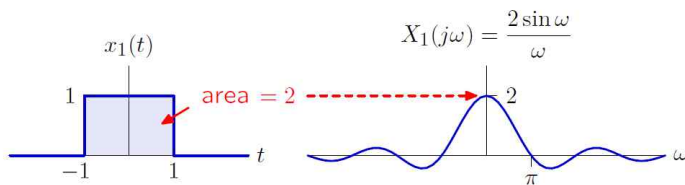
Fourier transform

$$X_1(j\omega) = \int_{-1}^1 e^{-j\omega t} dt = \left. \frac{e^{-j\omega t}}{-j\omega} \right|_{-1}^1 = \frac{2 \sin \omega}{\omega} \quad [\text{function of } \omega]$$

※ the value of $X(j\omega)$ at $\omega = 0$

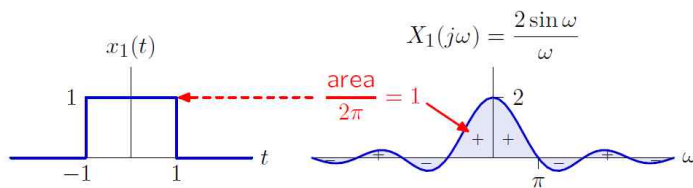
The value of $X(j\omega)$ at $\omega = 0$ is the integral of $x(t)$ over time t .

$$X(j\omega)|_{\omega=0} = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(t) e^{j0t} dt = \int_{-\infty}^{\infty} x(t) dt$$



The value of $x(0)$ is the integral of $X(j\omega)$ divided by 2π .

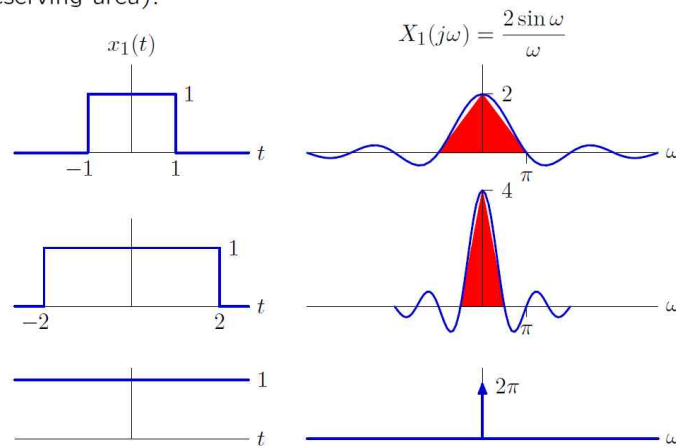
$$x(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) d\omega$$



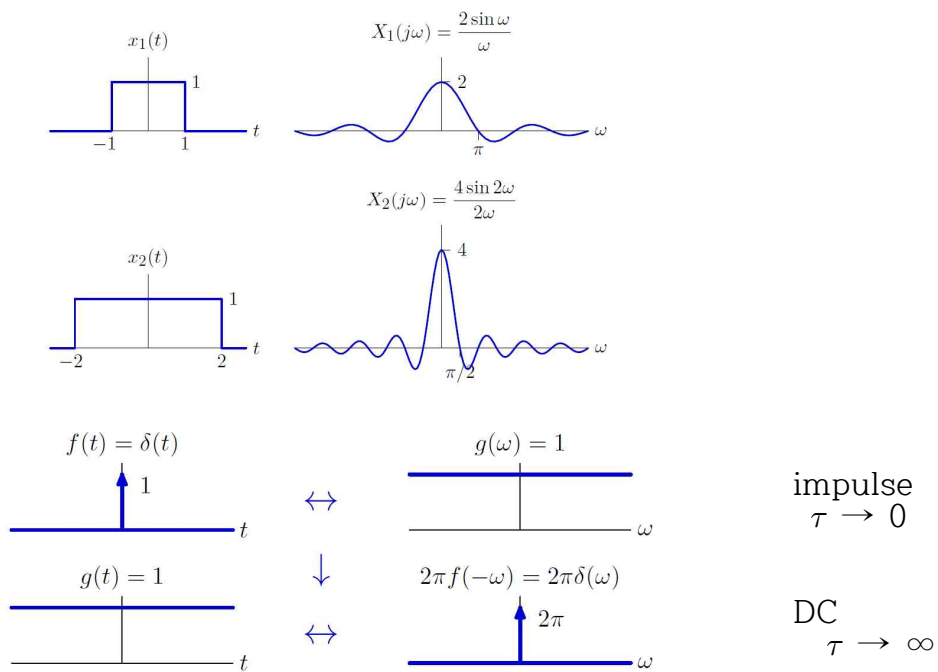
※ pulse width bigger and bigger ($\tau \rightarrow \infty$),

the frequency of first zero-crossing smaller and smaller.

Stretching time compresses frequency and increases amplitude (preserving area).



※ $\tau \rightarrow 0$ (impulse) or $\tau \rightarrow \infty$ (DC)

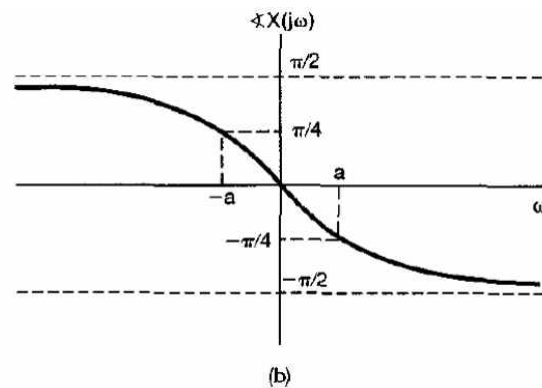
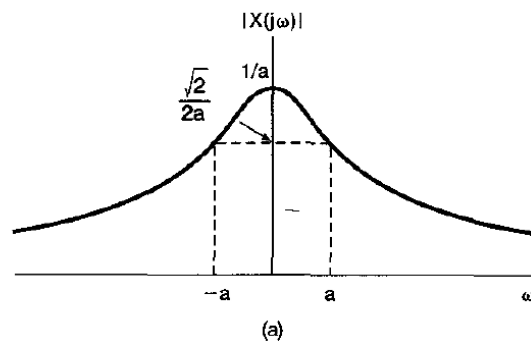


§4.1.3 Examples of CTFT

(Example 4.1)

$$x(t) = e^{-at} u(t) \quad a > 0$$

$$\begin{aligned} X(j\omega) &= \int_0^{\infty} e^{-at} e^{-j\omega t} dt = -\frac{1}{a+j\omega} e^{-(a+j\omega)t} \Big|_0^{\infty} \\ &= \frac{1}{a+j\omega}, \quad a > 0 \end{aligned}$$



(Example 4.2)

$$x(t) = e^{-a|t|} u(t) \quad a > 0$$

$$X(j\omega) = \int_{-\infty}^{\infty} e^{-a|t|} e^{-j\omega t} dt = \frac{2a}{a^2 + \omega^2}, \quad a > 0$$

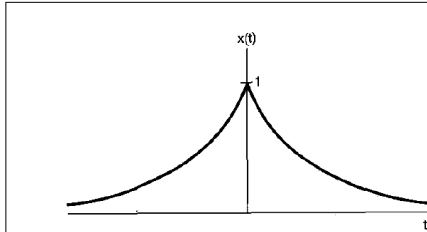
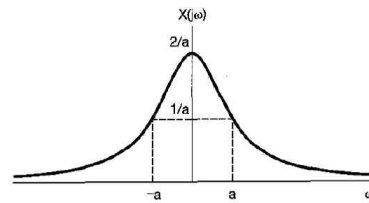
Figure 4.6 Signal $x(t) = e^{-a|t|}$ of Example 4.2.

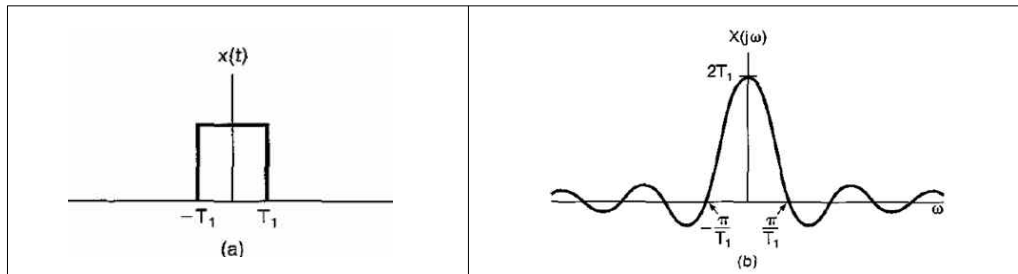
Figure 4.7 Fourier transform of the signal considered in Example 4.2 and depicted in Figure 4.6.

(Example 4.4) In Fig. 4.1, $T \rightarrow \infty$ and T_1 is fixed

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & |t| > T_1 \end{cases} \quad (4.16)$$

$$X(j\omega) = \int_{-T_1}^{T_1} e^{-j\omega t} dt = 2 \frac{\sin \omega T_1}{\omega} \quad (4.17)$$

Fig. 4.8 (a) and (b)

(Example 4.3) In Fig. 4.1, $T \rightarrow \infty$ and $T_1 \rightarrow 0$

$$x(t) = \delta(t)$$

$$X(j\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = 1$$

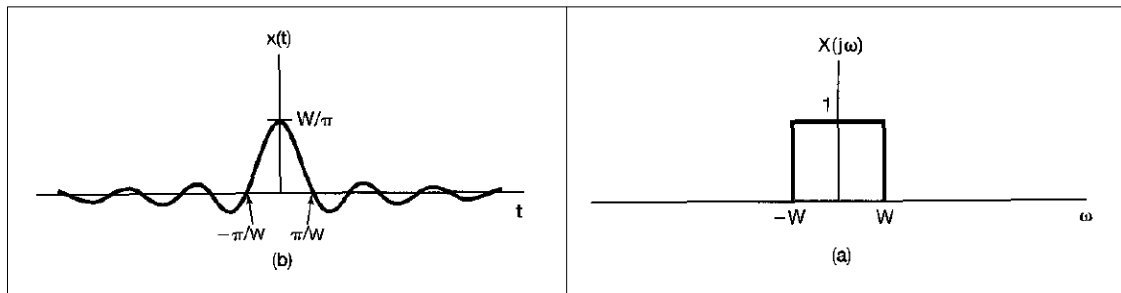
$$\ast X(j\omega) = 1 \text{ (meaning?)}$$

(Example 4.5)

$$X(j\omega) = \begin{cases} 1, & |\omega| < W \\ 0, & |\omega| > W \end{cases} \quad (4.18)$$

The inverse Fourier Transform

$$x(t) = \frac{1}{2\pi} \int_{-W}^W e^{j\omega t} d\omega = \frac{\sin Wt}{\pi t} \quad (4.19)$$

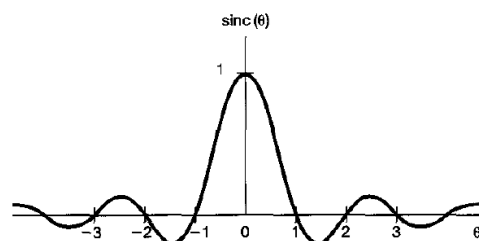


※ Duality

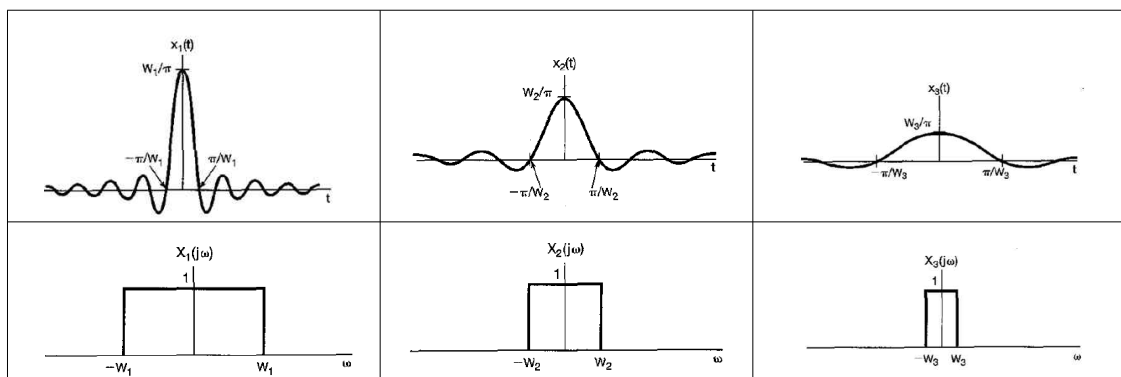
(Example 4.4), (Example 4.5)

● sinc function

$$\text{sinc}(\theta) = \frac{\sin(\pi\theta)}{\pi\theta} \quad (4.20)$$



Fourier Transform pair (Uncertainty Principle)



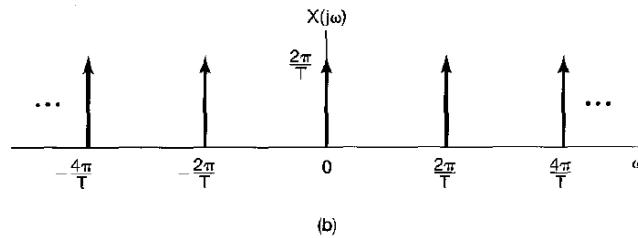
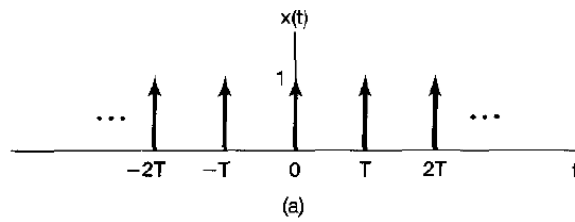
§4.2 The Fourier Transform for Periodic Signals

(Example 4.8) Impulse train

In Fig. 4.1, T is fixed and $T_1 \rightarrow 0$

$$x(t) = \sum_{k=-\infty}^{+\infty} \delta(t - kT)$$

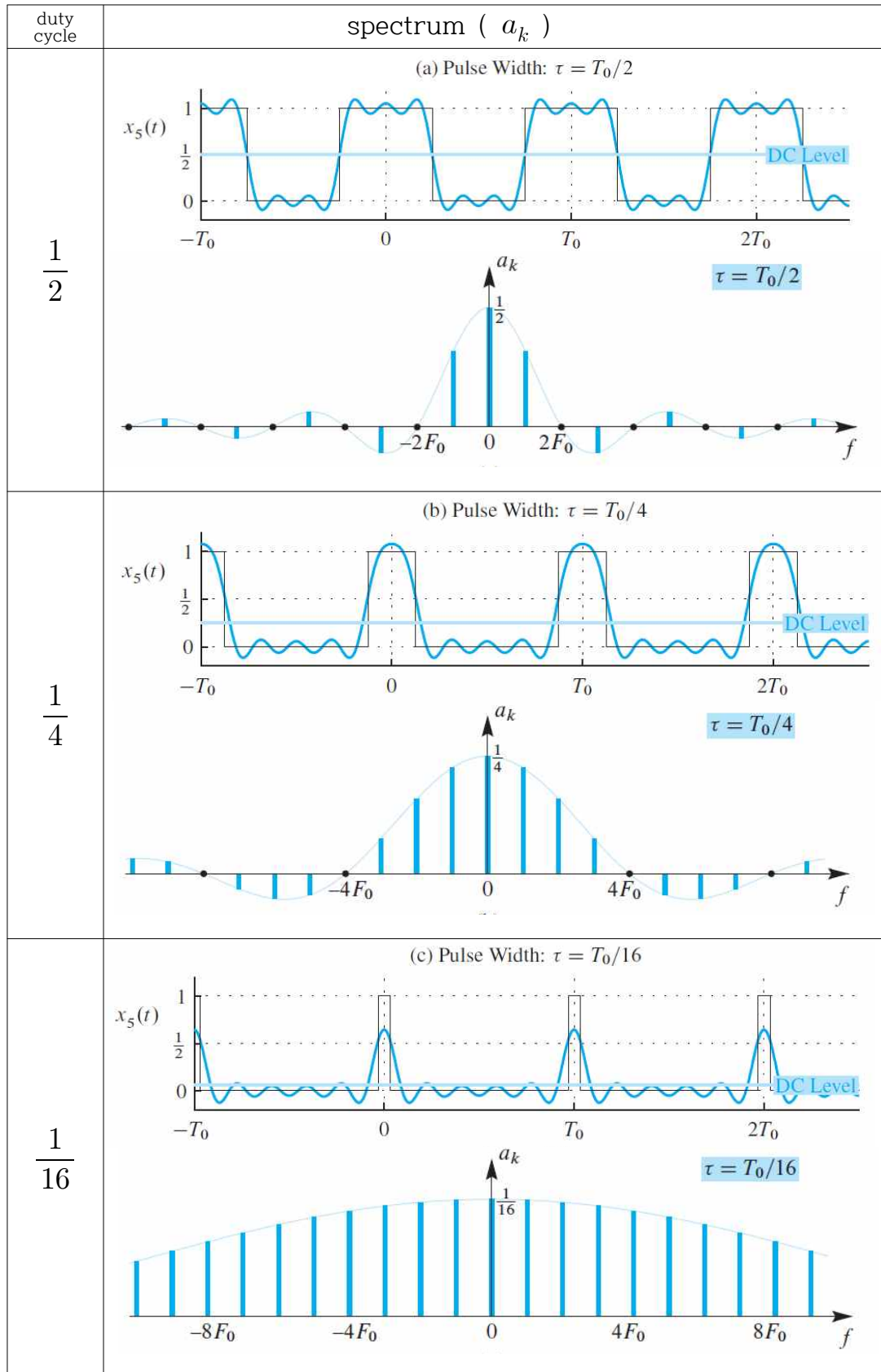
$$X(j\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{2\pi k}{T}\right)$$



※ If T is smaller, then the interval between spectra is bigger.

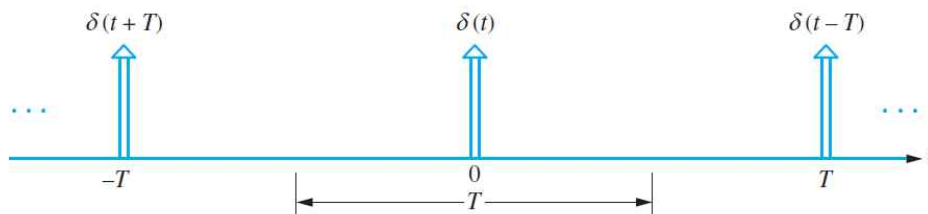
If T is bigger, then the interval between spectra is smaller.

(Chapter 3) CTFS (pulse width smaller, period fixed)



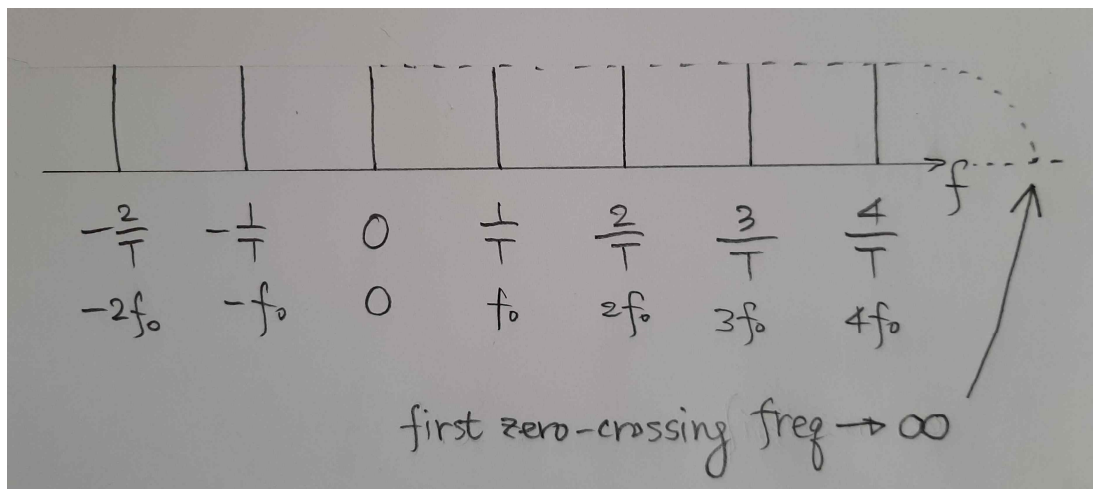
(Impulse Train) pulse with $(\tau) \rightarrow 0$

$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

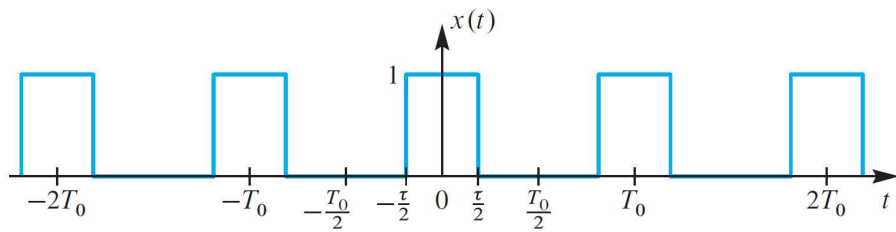


- Spectrum of a impulse train

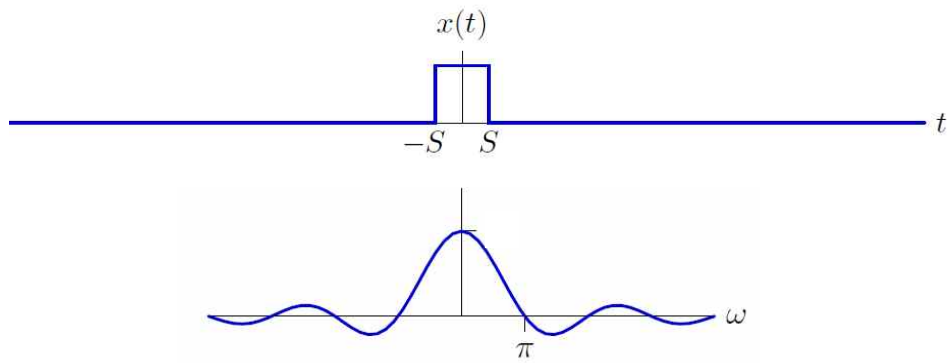
first zero-crossing frequency $(\frac{1}{\tau}) \rightarrow \infty$



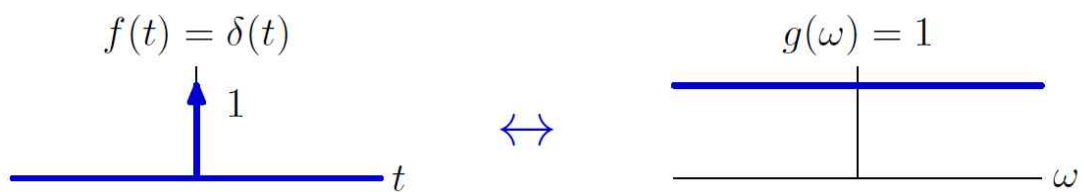
(Think about the spectrum)



(i) pulse width (τ) fixed, period (T_0) $\rightarrow \infty$ [nonperiodic]



(ii) pulse with (τ) $\rightarrow 0$, period (T_0) $\rightarrow \infty$ [impulse]



§4.3 Properties of the CTFT

§4.3.1 Linearity

$$ax(t) + by(t) \leftrightarrow aX(j\omega) + bY(j\omega) \quad (4.26)$$

§4.3.2 Time Shifting

$$x(t - t_0) \leftrightarrow e^{-j\omega t_0} X(j\omega) \quad (4.27)$$

same magnitude
different phase

§4.3.3 Conjugation and Conjugate Symmetry

$$x^*(t) \leftrightarrow X^*(-j\omega) \quad (4.28)$$

- Conjugate Symmetry

$$\text{If } x(t) \text{ is real, then } X(-j\omega) = X^*(j\omega). \quad (4.30)$$

§4.3.4 Differentiation and Integration

$$\frac{dx(t)}{dt} \leftrightarrow j\omega X(j\omega) \quad (4.31)$$

$$\int_{-\infty}^t x(\tau) d\tau \leftrightarrow \frac{1}{j\omega} X(j\omega) + \pi X(0) \delta(\omega) \quad (4.32)$$

§4.3.5 Time and Frequency Scaling ... skip

§4.3.6 Duality

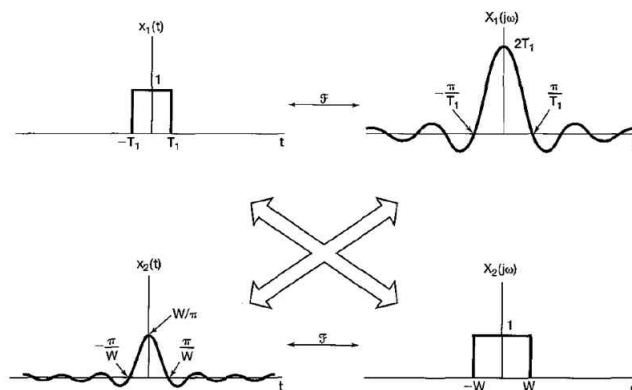


Figure 4.17 Relationship between the Fourier transform pairs of eqs. (4.36) and (4.37).

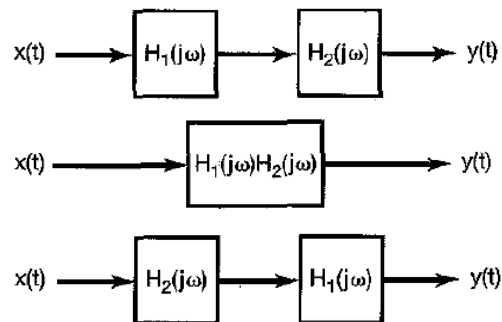
§4.3.7 Parseval's Relation

$$\int_{-\infty}^{+\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} |X(j\omega)|^2 d\omega \quad (4.43)$$

§4.4 The Convolution Property

$$y(t) = h(t) * x(t) \leftrightarrow Y(j\omega) = H(j\omega) X(j\omega) \quad (4.56)$$

※ Read pp.314-315 of the text.



(Example 4.15) time delay

$$h(t) = \delta(t - t_0)$$

$$H(j\omega) = e^{-j\omega t_0}$$

$$Y(j\omega) = H(j\omega) X(j\omega) = e^{-j\omega t_0} X(j\omega)$$

$$y(t) = x(t - t_0)$$

(Example 4.16) differentiation

$$y(t) = \frac{dx(t)}{dt}$$

$$Y(j\omega) = j\omega X(j\omega)$$

$$H(j\omega) = j\omega$$

(Example 4.17) integration

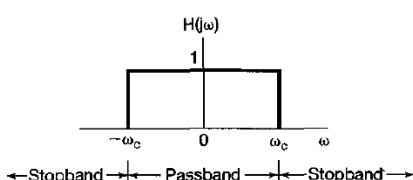
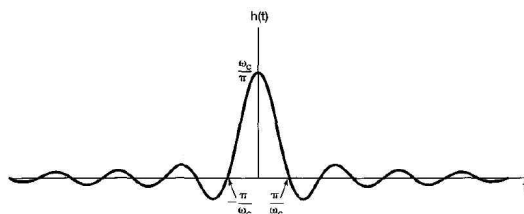
$$y(t) = \int_{-\infty}^t x(\tau) d\tau$$

$$H(j\omega) = \frac{1}{j\omega} + \pi \delta(\omega)$$

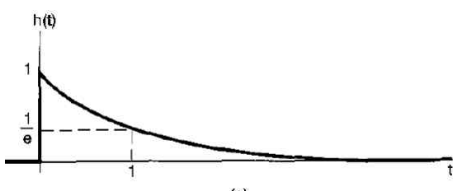
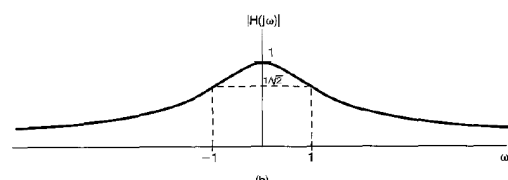
$$\begin{aligned} Y(j\omega) &= H(j\omega) X(j\omega) = \frac{1}{j\omega} X(j\omega) + \pi X(j\omega) \delta(\omega) \\ &= \frac{1}{j\omega} X(j\omega) + \pi X(0) \delta(\omega) \end{aligned}$$

(Example 4.18) frequency-selective filtering

(ideal low-pass filter [LPF])

$H(j\omega) = \begin{cases} 1, & \omega < \omega_c \\ 0, & \omega > \omega_c \end{cases}$	$h(t) = \frac{\sin \omega_c t}{\pi t}$
 Figure 4.20 Frequency response of an ideal lowpass filter.	 Figure 4.21 Impulse response of an ideal lowpass filter.

(Example) First-order LPF (one-pole LTI system)

$h(t) = e^{-t} u(t)$	$H(j\omega) = \frac{1}{j\omega + 1}$
 (a)	 (b)

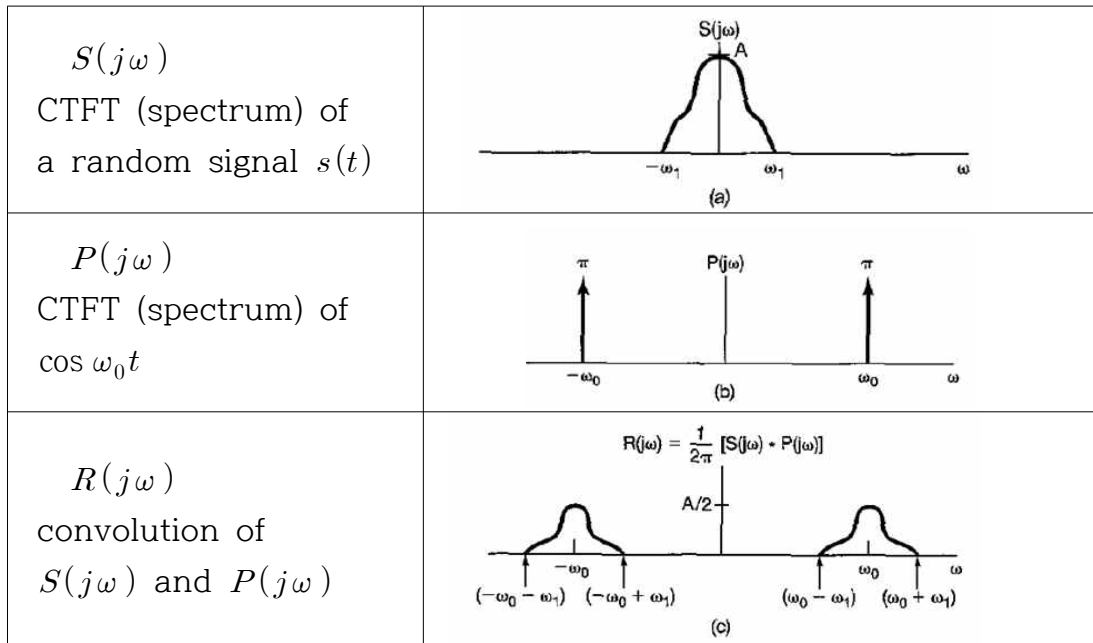
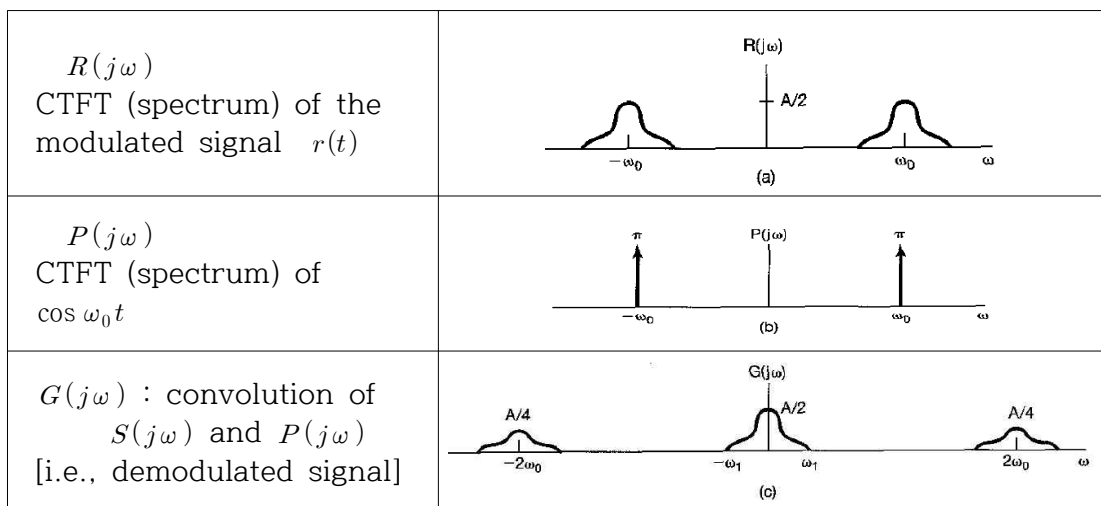
(Example 4.19)

$h(t) = e^{-at} u(t), \quad a > 0$	$H(j\omega) = \frac{1}{a + j\omega}$
$x(t) = e^{-bt} u(t), \quad b > 0$	$X(j\omega) = \frac{1}{b + j\omega}$
$y(t) = x(t) * h(t)$	$Y(j\omega) = X(j\omega)H(j\omega)$
Which one is easier to calculate, $y(t)$ [i.e., integration] or $Y(j\omega)$ [i.e., multiplication] ?	

§4.5 The Multiplication Property

$$s(t)p(t) \leftrightarrow \frac{1}{2\pi} \int_{-\infty}^{\infty} S(j\theta)P(j(\omega - \theta))d\theta \quad (4.70)$$

time domain $\leftrightarrow$ frequency domain	property	remarks
convolution $\leftrightarrow$ multiplication	convolution property	Eq.(4.56)
multiplication $\leftrightarrow$ convolution	modulation property	Eq.(4.70)

(Example 4.21) $r(t) = s(t)p(t)$ (Example 4.22) Demodulation $g(t) = r(t)p(t)$ 

● Frequency-selective filtering with variable center frequency

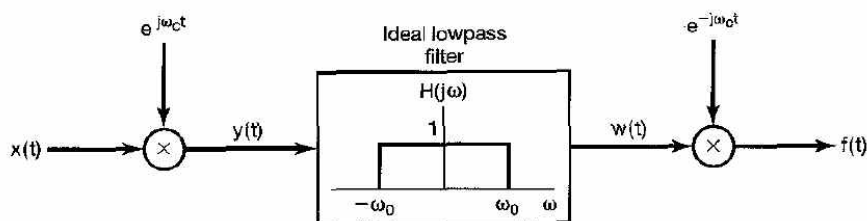


Figure 4.26 Implementation of a bandpass filter using amplitude modulation with a complex exponential carrier.

§4.6 Tables of CTFT Pairs

TABLE 4.1 PROPERTIES OF THE FOURIER TRANSFORM

Section	Property	Aperiodic signal	Fourier transform
		$x(t)$	$X(j\omega)$
		$y(t)$	$Y(j\omega)$
4.3.1	Linearity	$ax(t) + by(t)$	$aX(j\omega) + bY(j\omega)$
4.3.2	Time Shifting	$x(t - t_0)$	$e^{-j\omega t_0} X(j\omega)$
4.3.6	Frequency Shifting	$e^{j\omega_0 t} x(t)$	$X(j(\omega - \omega_0))$
4.3.3	Conjugation	$x^*(t)$	$X^*(-j\omega)$
4.3.5	Time Reversal	$x(-t)$	$X(-j\omega)$
4.3.5	Time and Frequency Scaling	$x(at)$	$\frac{1}{ a } X\left(\frac{j\omega}{a}\right)$
4.4	Convolution	$x(t) * y(t)$	$X(j\omega)Y(j\omega)$
4.5	Multiplication	$x(t)y(t)$	$\frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\theta)Y(j(\omega - \theta))d\theta$
4.3.4	Differentiation in Time	$\frac{d}{dt}x(t)$	$j\omega X(j\omega)$
4.3.4	Integration	$\int_{-\infty}^t x(t)dt$	$\frac{1}{j\omega} X(j\omega) + \pi X(0)\delta(\omega)$
4.3.6	Differentiation in Frequency	$tx(t)$	$j \frac{d}{d\omega} X(j\omega)$
4.3.3	Conjugate Symmetry for Real Signals	$x(t)$ real	$\begin{cases} X(j\omega) = X^*(-j\omega) \\ \Re\{X(j\omega)\} = \Re\{X(-j\omega)\} \\ \Im\{X(j\omega)\} = -\Im\{X(-j\omega)\} \\ X(j\omega) = X(-j\omega) \\ \angle X(j\omega) = -\angle X(-j\omega) \end{cases}$
4.3.3	Symmetry for Real and Even Signals	$x(t)$ real and even	$X(j\omega)$ real and even
4.3.3	Symmetry for Real and Odd Signals	$x(t)$ real and odd	$X(j\omega)$ purely imaginary and odd
4.3.3	Even-Odd Decomposition for Real Signals	$x_e(t) = \mathcal{E}\{x(t)\}$ [$x(t)$ real] $x_o(t) = \mathcal{O}\{x(t)\}$ [$x(t)$ real]	$\Re\{X(j\omega)\}$ $j\Im\{X(j\omega)\}$
4.3.7	Parseval's Relation for Aperiodic Signals		
	$\int_{-\infty}^{+\infty} x(t) ^2 dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) ^2 d\omega$		

TABLE 4.2 BASIC FOURIER TRANSFORM PAIRS

Signal	Fourier transform	Fourier series coefficients (if periodic)
$\sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$	$2\pi \sum_{k=-\infty}^{+\infty} a_k \delta(\omega - k\omega_0)$	a_k
$e^{j\omega_0 t}$	$2\pi \delta(\omega - \omega_0)$	$a_1 = 1$ $a_k = 0$, otherwise
$\cos \omega_0 t$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	$a_1 = a_{-1} = \frac{1}{2}$ $a_k = 0$, otherwise
$\sin \omega_0 t$	$\frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$	$a_1 = -a_{-1} = \frac{1}{2j}$ $a_k = 0$, otherwise
$x(t) = 1$	$2\pi \delta(\omega)$	$a_0 = 1$, $a_k = 0$, $k \neq 0$ (this is the Fourier series representation for any choice of $T > 0$)
Periodic square wave		
$x(t) = \begin{cases} 1, & t < T_1 \\ 0, & T_1 < t \leq \frac{T}{2} \end{cases}$ and $x(t + T) = x(t)$	$\sum_{k=-\infty}^{+\infty} \frac{2 \sin k\omega_0 T_1}{k} \delta(\omega - k\omega_0)$	$\frac{\omega_0 T_1}{\pi} \operatorname{sinc}\left(\frac{k\omega_0 T_1}{\pi}\right) = \frac{\sin k\omega_0 T_1}{k\pi}$
$\sum_{n=-\infty}^{+\infty} \delta(t - nT)$	$\frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{2\pi k}{T}\right)$	$a_k = \frac{1}{T}$ for all k
$x(t) = \begin{cases} 1, & t < T_1 \\ 0, & t > T_1 \end{cases}$	$\frac{2 \sin \omega T_1}{\omega}$	—
$\frac{\sin Wt}{\pi t}$	$X(j\omega) = \begin{cases} 1, & \omega < W \\ 0, & \omega > W \end{cases}$	—
$\delta(t)$	1	—
$u(t)$	$\frac{1}{j\omega} + \pi \delta(\omega)$	—
$\delta(t - t_0)$	$e^{-j\omega t_0}$	—
$e^{-at} u(t)$, $\Re\{a\} > 0$	$\frac{1}{a + j\omega}$	—
$te^{-at} u(t)$, $\Re\{a\} > 0$	$\frac{1}{(a + j\omega)^2}$	—
$\frac{t^{n-1}}{(n-1)!} e^{-at} u(t)$, $\Re\{a\} > 0$	$\frac{1}{(a + j\omega)^n}$	—

§4.7 Systems Characterized by LCCDE

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k} \quad (4.72)$$

Fourier transforming both sides

$$\sum_{k=0}^N a_k (j\omega)^k Y(j\omega) = \sum_{k=0}^M b_k (j\omega)^k X(j\omega)$$

Therefore,

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{\sum_{k=0}^M b_k (j\omega)^k}{\sum_{k=0}^N a_k (j\omega)^k} \quad (4.76)$$

※ $H(j\omega)$ is the Frequency Response of the LTI system.

(Example 4.24)

$$\frac{dy(t)}{dt} + ay(t) = x(t), \quad a > 0$$

Therefore,

$$H(j\omega) = \frac{1}{j\omega + a}$$

The inverse Fourier transform

$$h(t) = e^{-at} u(t)$$

(Example 4.25) and (Example 4.26)

4.8 Summary